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# MANGANESE SULPHATE MONOHYDRATE POWDER

| | | | |
|---|---|---|---|
| **Version** | V1301 | **Origin** | China |
| **Molecular Weight** | 169.01 | **Packaging** | 50lb, 55.12lb, 2000lb, 2204.6lb |
| **CAS - No.** | 10034-96-5 | **Storage** | cool, clean and dry |
| **Chemical formula** | $MnSO_4 \cdot H_2O$ | **Shelf life** | 24 months |

***Standard and Directive Reference:**

HG/T 2936-1999 (feed grade), AFIA, AAFCO, ANAC, AAFC, 2202/32/EC, 2005/87/EC, 2006/13/EC, CLP regulation, GB10648-1999

***Application:** it is intended for manufacturing animal feed supplements or for agricultural use for plant nutrition and industrial use.

## Typical chemical analysis

**Content:** min.31.8% Mn

**Sulphur:** min.17.5% S

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**Heavy metal content:**

| | |
|---|---|
| Arsenic: | max. 5 mg/kg |
| Lead: | max. 10 mg/kg |
| Cadmium: | max. 10mg/kg |
| Mercury: | max. 0.5mg/kg |

## Physical Analysis:

| | |
|---|---|
| Flow | Free flow; dust free |
| Appearance | white to pinkish crystalline powder |
| Bulk density | 1400kg/m3 |
| Solubility in water | 99.95% |
| Particle Size | 95%min pass through 60 mesh |

## Packaging:

Coated woven polypropylene 25kg/ 1 ton bag with inner liner

Wooden pallets meet ISPM 15 regulations

Pallets are stretched wrapped.

Special packaging available on request.

## Label

Label includes batch number, net weight, manufacturing & expire dates.

Neutral label or customer label are available on request.

## Safety and storage conditions

Store under clean, dry conditions

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# MATERIAL SAFETY DATA SHEET

# MANGANESE SULPHATE MONOHYDRATE

## SECTION 1: PRODUCT AND PRODUCER/SUPPLIER DETAILS

PRODUCT NAME: Manganese Sulfate Monohydrate

SYNONYMS, TRADE NAMES: Manganese Sulfate

SUPPLIER: P ROMISING CHEMICALS CO., LTD

ROO M 1310, DONGYI INTERNATIONAL BUILDING, 488 N ORTH SHAOSHAN ROA D

CHANGSHA HUNAN CHINA. POSTAL CODE: 410007

TEL: 0086 731 85328848 FAX: 0086 731 85328847

## SECTION 2 - CHEMICAL NAME & INFORMATION ON INGREDIENTS

**Chemical Name:** Manganese Sulfate (CAS : 7785-87-7) (EINECS: 232-089-9)

**R-Phrases:** R36/38- Irritating to eyes and skin.

R50/53- Very toxic to aquatic organisms, may cause long term adverse effects in the aquatic environment.

**Formula:** $MnSO_4.H_2O$

**Synonyms:** Dried Manganese Sulphate, Manganous Sulphate

**% Range:** 98 - 100%

Trace materials normally not exceeding 0.5%

There are no physically added chemical substances. The impurities are inherent to the raw materials used in the production of Manganese Sulfate.

## SECTION 3 - PHYSICAL & CHEMICAL CHARACTERISTICS

**Melting Point:** Above 730°C

**Specific Gravity:** 3.70 g/cm³ at 20°C

**Solubility in water:** Soluble in water.

**Formula Wt.** 179.40

**Appearance & Odour:** White to pinkish crystalline powder, odourless

## SECTION 4 - HAZARDS IDENTIFICATION

**Physical / chemical hazards :** Harmful if swallowed, causes irritation, avoid contact with eyes and skin.

**Environmental hazard :** R51/53 - Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

**Adverse human health effects**

**- Inhalation :** Causes irritation to the respiratory tract.

**- Skin contact :** Skin contact may cause irritation.

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- **Eye contact :** Excessive dust exposure may cause eye irritation.
- **Ingestion :** Health hazard if ingested and can cause abdominal pain, nausea, diarrhea, cramps, and damage to gastrointestinal tract.

## SECTION 5 – FIRE, FIGHTING & EXPLOSION DATA

**Fire:** Non Combustible.

**Flammability:** Non Flammable

**Flash point:** Not combustible

**Extinguishing media:** In case of fire in immediate vicinity use any extinguishing agent appropriate for surrounding area.

**Protection of fire-fighters:** Use breathing apparatus (self-contained breathing apparatus with full face shield). Wear suitable protective clothing.

**Hazardous combustion and** May emit Sox fumes.

## SECTION 6 - STABILITY & REACTIVITY DATA

**Stability:** Stable under ordinary conditions.

**Hazardous Reactions:** Avoid Heating! Decomposition can emit Sox fumes

## SECTION 7 – Toxicological Information

ZARD STATEMENT: H373
PRECAUTIOANRY STATEMENT PREVENTION P260
RECAUTIONARY STATEMENT RESPONSE (ORAL): P314
PRECAUTIONARY STATEMENT DISPOSAL (ORAL): P501
HAZARD STATEMENT: H411: TOXIC TO AQUATIC LIFE WIT LONG LASTING EFFECTS
PRECAUTIOANRY STATEMENT PREVENTION : P273
PRECAUTIONARY STATEMENT RESPONSE : P391
PRECAUTIONARY STATEMENT DISPOSAL: P501
HAZARD STATEMENT: H410: VERY TOXIC TO AQUATIC LIFE WITH LONG LASTING EFFECTS
PRECAUTIOANRY STATEMENT PREVENTION : P273
PRECAUTIONARY STATEMENT RESPONSE : P391
PRECAUTIONARY STATEMENT DISPOSAL: P501

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## SECTION 8 - FIRST AID MEASURES

**Effects and symptoms:** No specific effects and/or symptoms have been reported or known.

**First aid**

- **Skin contact:** Rinse immediately with plenty of water
- **Inhalation:** Remove victim to fresh air. If feeling unwell, immediately seek medical attention.
- **Eye contact:** Rinse immediately with clean water for 15 minutes. If feeling unwell, immediately seek medical attention.

**- Ingestion:** Immediately rinse mouth with water. Do not induce vomiting. Give water to drink. If feeling unwell, immediately seek medical attention.

**Protection of rescue personnel:** Avoid all unnecessary exposure. Use appropriate protection (seeSection 9).

## SECTION 9 - PERSONAL PROTECTION & EXPOSURE CONTROLS

**- Respiratory protection:** Respiratory protection equipment not absolutely necessary.

**- Hand protection:** Gloves. Material: leather gloves, cotton gloves, rubber gloves.

**- Skin protection:** Work clothes.

**- Eye protection:** Chemical goggles or safety glasses.

**Occupational Exposure Limits**

**8 Hour TLV-TWA [mg/m3] :** 10 (Dust) 5 (fume)

**Engineering measures:** Mechanical ventilation is recommended.

## SECTION 10 ACCIDENTAL RELEASE MEASURES

**Personal precautions:** Wear suitable gloves and eye/face protection. Avoid dust creation.

**Environmental precautions:** Do not let product enter drains, sewage system, ground water.

**Methods for cleaning up:** Sweep up and shovel into suitable containers for disposal. Re-use or recycle waste.

## SECTION 11 - HANDLING AND STORAGE

**Handling & storing:** Good housekeeping - store in a cool dry place - keep containers shut. Avoid making dust.

**Packing Materials:** All materials suitable. Dry Storage with tightly closed lids.

## SECTION 12 - ECOLOGICAL INFORMATION

**Ecotoxicity:** Water soluble, toxic to aquatic life.

**Fish Toxicity**: LC50/96hr/cichlid 13 ppm.

**Mammalian Toxicity:** LD50/oral/mouse 57 mg/kg
LD50/oral/rat 2900 mg/kg.

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## SECTION 13 - DISPOSAL CONSIDERATIONS

**Disposal:** Comply with local regulations for disposal.

**Waste of residues:** S60 - This material and its container must be disposed of as hazardous waste. Because of possible pollution, remove as industrial waste or hazardous waste.
S61- Avoid release to the environment. Refer to special instructions/safety data sheets.

**Contaminated packaging:** Keep waste packaging separate.
Because of possible pollution, remove as industrial waste or hazardous waste.

## SECTION 14 - TRANSPORT INFORMATION

**As Common goods transportation**

**H.S. Code:** 2833.00

## SECTION 15 - REGULATORY INFORMATION

**R Phrase(s):** R36/38 – Irritating to eyes and skin.

R51/53 - Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

**S Phrase(s):** S2 - Keep out of reach of children. S22- Do not breath dust.

S24- Avoid contact with skin. S25 Avoid contact with eyes.

S60 – This material and/or its container must be disposed of as hazardous waste.

S61 - Avoid release to the environment. Refer to special instructions/Safety datasheets.

## SECTION 16 - OTHER INFORMATION

**Date of Issue:** Dec. 2011

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**MSDS prepared by:** Liu Yi

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